

		$= 3.54 \text{ m}^3$
8	B	Extension for spring X, $x_1 = \frac{F}{k_1} = \frac{1.5}{5.0} = 0.30 \text{ m}$ Extension for spring Y, $x_2 = \frac{F}{k_2} = \frac{1.5}{10.0} = 0.15 \text{ m}$ Elastic P.E. stored $= \frac{1}{2} kx^2$ Total elastic P.E. stored $= \frac{1}{2} (5)(0.3)^2 + \frac{1}{2} (10)(0.15)^2 \text{ J} = 0.34 \text{ J}$
9	C	Loss in G.P.E. by Y $\equiv$ gain in G.P.E. by X + gain in K.E. by X and Y